

(9~10 学时)

2.3 支路电流法

2.4 网孔电流法与回路电流法

作业: 2.11(a), 2.13, 2.14, 2.17

2.11(a) 2.13 2.14 2.17

2.21 2.22 2.24(b) 2.26 2.29

(11~12 学时)

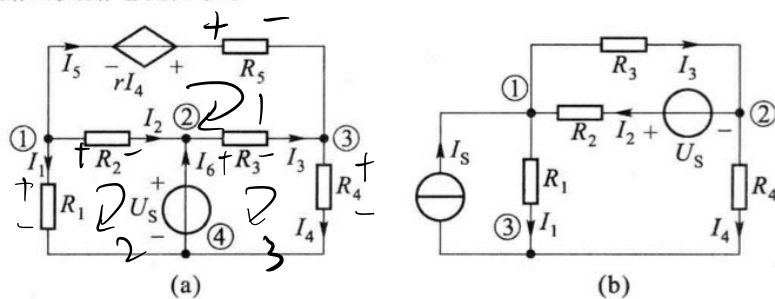
2.5 节点电压法

2.6 运算放大器

2.7 含运算放大器电路的分析

作业: 2.21, 2.22, 2.24(b), 2.26, 2.29

2.11 列写图示电路的支路电流方程。



图题 2.11

2.11(a) ①: $I_1 = I_2 + I_5$

②: $I_3 = I_2 + I_6$

③: $I_4 = I_3 + I_5$

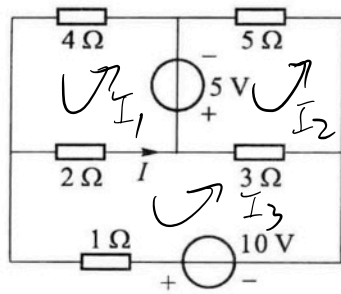
1: $rI_4 - I_5R_5 - I_3R_3 - I_2R_2 = 0$

2: $I_1R_1 - I_2R_2 - U_5 = 0$

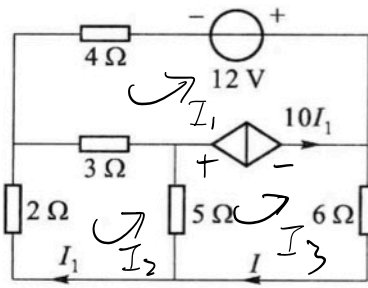
3: $U_5 - I_3R_3 - I_4R_4 = 0$

2.13 用回路电流法求图示电路的电流 I 。

2.14 用回路电流法求图示电路的电流 I 。



图题 2.13



图题 2.14

$$2.13 \begin{cases} 6I_1 - 2I_3 = -5V \\ 8I_2 - 3I_3 = 5V \\ 6I_3 - 2I_1 - 3I_2 = -10V \end{cases}$$

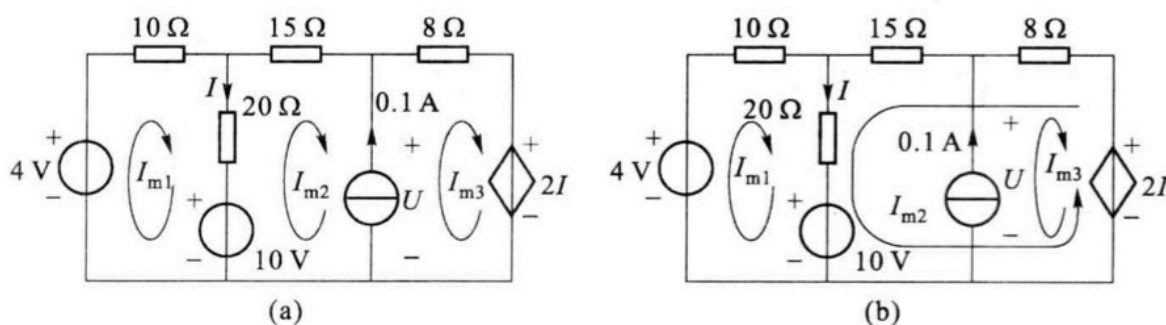
$$I = I_1 - I_3$$

$$\text{解得: } I = 0.72A$$

$$2.14 \begin{cases} 7I_1 - 3I_2 = -12V - U \\ 10I_2 - 5I_3 - 3I_1 = 0 \\ 11I_3 - 5I_2 = U \\ I_3 = -I \\ I_2 = -I_1 \\ I_1 - I_3 = 10I_1 \end{cases}$$

$$\text{解得: } I = 5A$$

2.17 图示电路,分别按图(a)、(b)规定的回路列出回路电流方程。

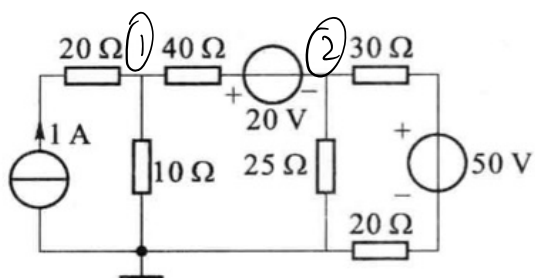


图题 2.17

$$\begin{aligned}
 (a) \quad & \begin{cases} 30 I_{m1} - 20 I_{m2} = -6V \\ 35 I_{m2} - 20 I_{m1} = 10V - U \\ 8 I_{m3} = U - 2I \\ I_{m3} - I_{m2} = 0.1A \\ I = I_{m1} - I_{m2} \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \begin{cases} 30 I_{m1} + 20 I_{m2} = -6V \\ 43 I_{m2} + 20 I_{m1} = -10V + 2I \\ 8 I_{m3} = U - 2I \\ I_{m3} = 0.1A \\ I = I_{m1} + I_{m2} \end{cases}
 \end{aligned}$$

2.21



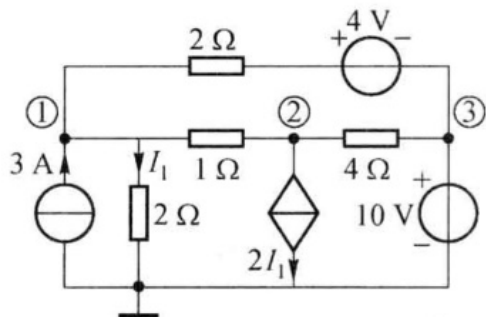
图题 2.21

$$\begin{aligned}
 ①: & \left(\frac{1}{40} + \frac{1}{10} \right) U_1 - \frac{1}{40} U_2 = 1A + \frac{20V}{40\Omega} \\
 ②: & \left(\frac{1}{40} + \frac{1}{50} + \frac{1}{25} \right) U_2 - \frac{1}{40} U_1 = \frac{50V}{50\Omega} - \frac{20V}{40\Omega}
 \end{aligned}$$

$$\therefore U_1 = 14V \quad U_2 = 10V$$

$$P = (U_1 + 20\Omega \times 1A) \times 1A = 34W$$

2.22



图题 2.22

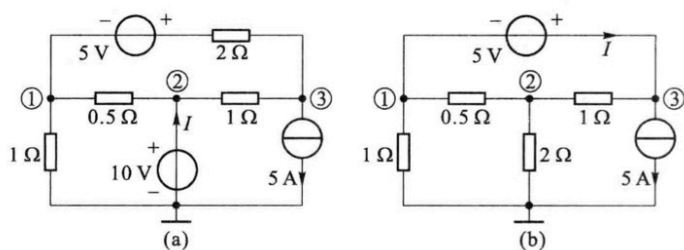
$$U_3 = 0 + 10V = 10V$$

$$\textcircled{1}: \left(\frac{1}{2} + \frac{1}{2} + 1\right)U_1 - U_2 - \frac{1}{2}U_3 = 3A + \frac{4V}{2\Omega}$$

$$\textcircled{2}: \left(1 + \frac{1}{4}\right)U_2 - U_1 - \frac{1}{4}U_3 = -2I_1$$

$$\text{解得: } \begin{cases} U_1 = 6V \\ U_2 = 2V \\ U_3 = 10V \end{cases}$$

2.24



图题 2.24

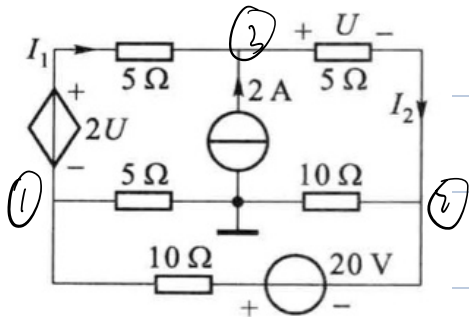
$$(b) \textcircled{1}: (1+2)U_1 - 2U_2 = -I$$

$$\textcircled{2}: \left(2 + 1 + \frac{1}{2}\right)U_2 - 2U_1 - U_3 = 0$$

$$\textcircled{3}: U_3 - U_2 = -5 + I$$

$$U_3 - U_1 = 5V$$

$$\text{解得: } \begin{cases} U_1 = -4V \\ U_2 = -2V \\ U_3 = 1V \\ I = 8A \end{cases}$$



图题 2.26

$$2.26 \quad ①: \left(\frac{1}{5} + \frac{1}{5} + \frac{1}{10}\right)U_1 - \frac{1}{5}U_3 - \frac{1}{10}U_2 = \frac{20}{10} - \frac{2V}{5}$$

$$②: \left(\frac{1}{5} + \frac{1}{10} + \frac{1}{10}\right)U_2 - \frac{1}{5}U_3 - \frac{1}{10}U_1 = -2$$

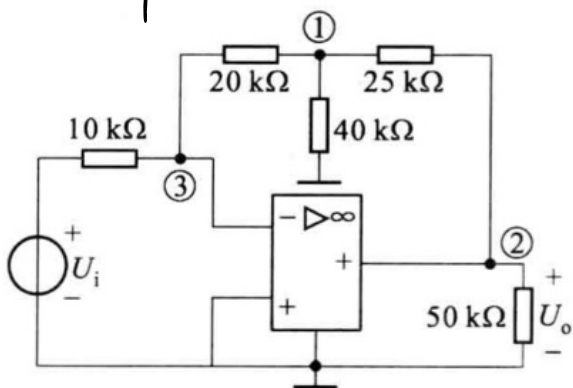
$$③: \left(\frac{1}{5} + \frac{1}{5}\right)U_3 - \frac{1}{5}U_1 - \frac{1}{5}U_2 = 2 + \frac{2V}{5}$$

$$U_3 - U_2 = U \quad U = 5I_2 \quad I_1 = \frac{U_3 - U_1 - 2V}{5}$$

解得:
$$\begin{cases} U_1 = \frac{10}{3}V \\ U_2 = \frac{40}{3}V \\ U_3 = 35V \end{cases}$$

$$\begin{cases} I_1 = \frac{7}{3}A \\ I_2 = \frac{13}{3}A \end{cases}$$

2.29



图题 2.29

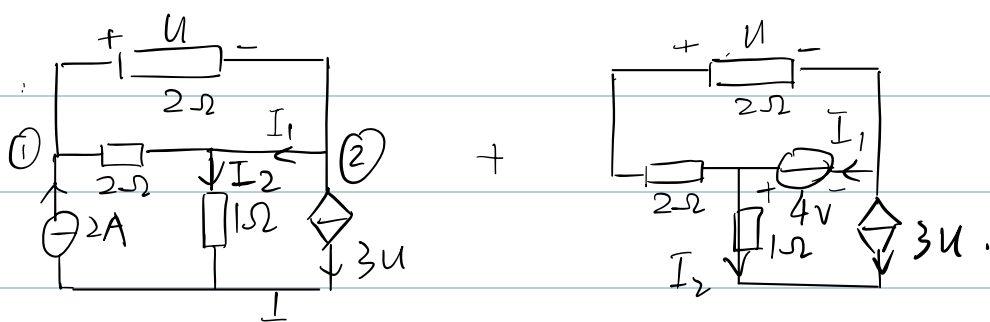
$$①: \left(\frac{1}{20} + \frac{1}{25} + \frac{1}{40}\right)U_1 - \frac{1}{20}U_3 - \frac{1}{20}U_2 = 0$$

$$③: \left(\frac{1}{10} + \frac{1}{20}\right)U_3 - \frac{1}{20}U_1 = \frac{U_i}{10}$$

$U_2 = U_o$

解得:
$$\frac{U_o}{U_i} = -\frac{23}{4}$$

(b)



$$\begin{cases} (\frac{1}{2} + \frac{1}{2}) u_1 - (\frac{1}{2} + \frac{1}{2}) u_2 = 2 \\ (\frac{1}{2} + \frac{1}{2}) u_2 - (\frac{1}{2} + \frac{1}{2}) u_1 = -3u \\ u_1 - u_2 = u \end{cases}$$

$$u = 2V$$

$$I_2 = -6A$$

$$I_1 = I_2 + 1A = -5A$$

$$\therefore \begin{cases} u_1 = -2V \\ u_2 = -4V \\ u = 2V \end{cases}$$

$$\therefore \begin{cases} I_1 = \frac{u}{2} - 3u = -5A \\ I_2 = -4A \end{cases}$$

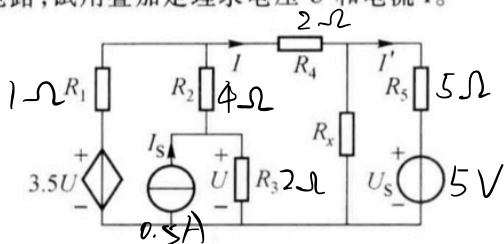
$$\therefore I = -5A - 5A = -10A$$

$$I_{1\Omega} = -4A - 6A = -10A$$

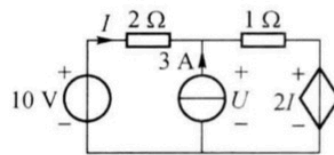
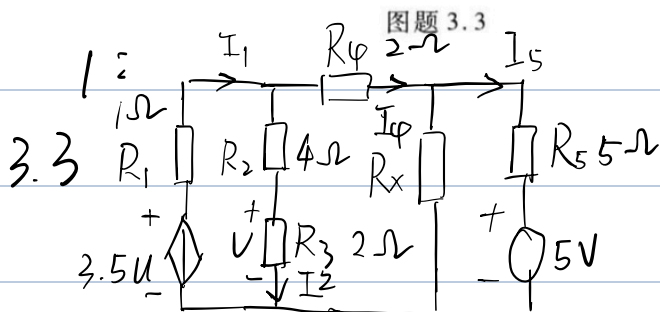
$$P = I_{1\Omega}^2 \cdot 1\Omega = 100W$$

3.3 图示电路中,已知 $R_1 = 1\Omega$, $R_2 = 4\Omega$, $R_3 = R_4 = 2\Omega$, $R_5 = 5\Omega$, $I_s = 0.5A$, $U_s = 5V$ 。欲使 U_s 中的电流为零,求 R_x 的值。

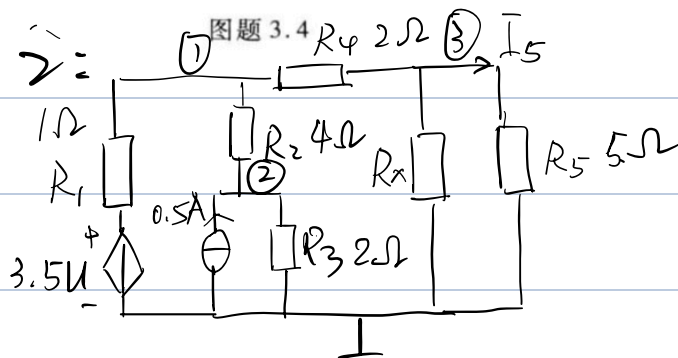
3.4 图示电路,试用叠加定理求电压 U 和电流 I 。



图题 3.3



图题 3.4



$$1: I_2 = \frac{V}{2} \quad R_2, R_3 \text{ 压降 } V + 2V = 3V$$

$$\therefore U_{R_1} = 3.5V - 3V = 0.5V$$

$$I_1 = \frac{0.5V}{1} = \frac{1}{2}V$$

$$\therefore I_\varphi = 0 \quad I_5 = -\frac{1}{5+R_x} \cdot 5$$

$$2: \begin{cases} (1 + \frac{1}{2} + \frac{1}{\varphi}) U_1 - \frac{1}{\varphi} U_2 - \frac{1}{2} U_3 = 3.5V \\ (\frac{1}{2} + \frac{1}{\varphi}) U_2 - \frac{1}{\varphi} U_1 = 0.5 \\ (\frac{1}{5} + \frac{1}{2} + \frac{1}{R_x}) U_3 - \frac{1}{2} U_1 = 0 \end{cases}$$

$$\begin{cases} (\frac{1}{2} + \frac{1}{\varphi}) U_2 - \frac{1}{\varphi} U_1 = 0.5 \\ (\frac{1}{5} + \frac{1}{2} + \frac{1}{R_x}) U_3 - \frac{1}{2} U_1 = 0 \end{cases}$$

$$U_2 = U$$

$$I_5 = \frac{1}{5} U_3$$

$$\text{解得: } \begin{cases} U_1 = \frac{25R_x}{10+2R_x} + 5 \\ U_2 = \frac{25R_x}{30+6R_x} + \frac{7}{3} \\ U_3 = \frac{25R_x}{10+2R_x} \end{cases}$$

$$\therefore I_5 = \frac{5R_x}{10+2R_x}$$

$$\therefore -\frac{5}{5+R_x} + \frac{5R_x}{10+2R_x} = 0 \quad \text{解得: } R_x = 2\Omega$$

$$7U_1 - 15U_2 = 2U_3$$

$$3U_2 - U_1 = 2$$

$$2U_1 = 2U_3 + 10$$

$$U_1 = U_3 + 5$$

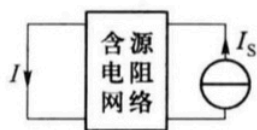
$$(\frac{7}{10} + \frac{1}{R_x}) U_3 = \frac{1}{2} U_3 + \frac{5}{2}$$

$$(\frac{7}{5} + \frac{2}{R_x}) U_3 = U_3 + 5$$

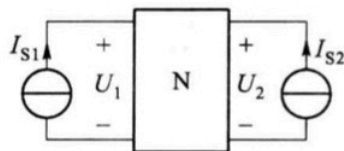
$$(\frac{2}{5} + \frac{2}{R_x}) U_3 = 5$$

$$U_3 = \frac{5 \cdot 5R_x}{2(5+R_x)}$$

$$\frac{1}{3} U_1 + \frac{2}{3}$$



图题 3.5



图题 3.6

3.6 图示电路中, N 为无独立源二端口网络。(1) 当 $I_{S1} = 2\text{ A}$, $I_{S2} = 0$ 时, I_{S1} 输出功率为 28 W , 且 $U_2 = 8\text{ V}$; (2) 当 $I_{S1} = 0$, $I_{S2} = 3\text{ A}$ 时, I_{S2} 的输出功率为 54 W , 且 $U_1 = 12\text{ V}$ 。求当 $I_{S1} = 2\text{ A}$, $I_{S2} = 3\text{ A}$ 共同作用时每个电流源的输出功率。

$$3.6 \quad U_1 = a I_{S1} + b I_{S2}$$

$$U_2 = c I_{S1} + d I_{S2}$$

$$(1) : U_1 I_{S1} = 28\text{ W} \quad U_1 = \frac{28\text{ W}}{2\text{ A}} = 14\text{ V}$$

$$(2) : U_2 I_{S2} = 54\text{ W} \quad U_2 = \frac{54\text{ W}}{3\text{ A}} = 18\text{ V}$$

$$\therefore \begin{cases} 14 = 2a \\ 8 = 2c \\ 12 = 3b \\ 18 = 3d \end{cases}$$

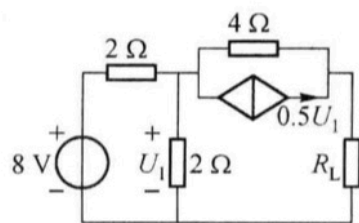
$$\therefore \begin{cases} a = 7 \\ b = 4 \\ c = 4 \\ d = 6 \end{cases}$$

$$\therefore \begin{cases} U_1 = 7 I_{S1} + 4 I_{S2} = 14 + 12 = 26\text{ (V)} \\ U_2 = 4 I_{S1} + 6 I_{S2} = 8 + 18 = 26\text{ (V)} \end{cases}$$

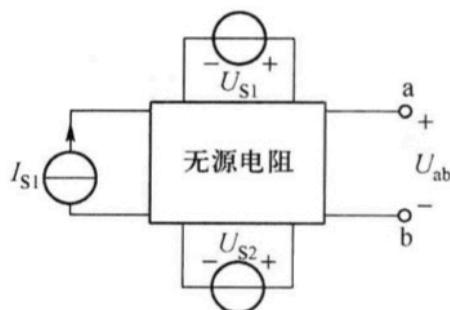
$$\therefore P_1 = U_1 I_{S1} = 52\text{ W}$$

$$P_2 = U_2 I_{S2} = 78\text{ W}$$

3.14 图示电路中,当电流源 I_{S1} 和电压源 U_{S1} 反向时(U_{S2} 不变),电压 U_{ab} 是原来的 0.5 倍;当 I_{S1} 和 U_{S2} 反向时(U_{S1} 不变),电压 U_{ab} 是原来的 0.3 倍。问:仅 I_{S1} 反向时(U_{S1}, U_{S2} 均不变),电压 U_{ab} 应为原来的几倍?



图题 3.13



图题 3.14

$$u = aU_{S1} + bU_{S2} + cI_{S1}$$

$$\begin{cases} u = aU_{S1} + bU_{S2} + cI_{S1} \\ 0.5u = -aU_{S1} + bU_{S2} - cI_{S1} \\ 0.3u = aU_{S1} - bU_{S2} - cI_{S1} \end{cases}$$

$$\therefore \begin{cases} aU_{S1} = 0.65u \\ bU_{S2} = 0.75u \\ cI_{S1} = -0.4u \end{cases}$$

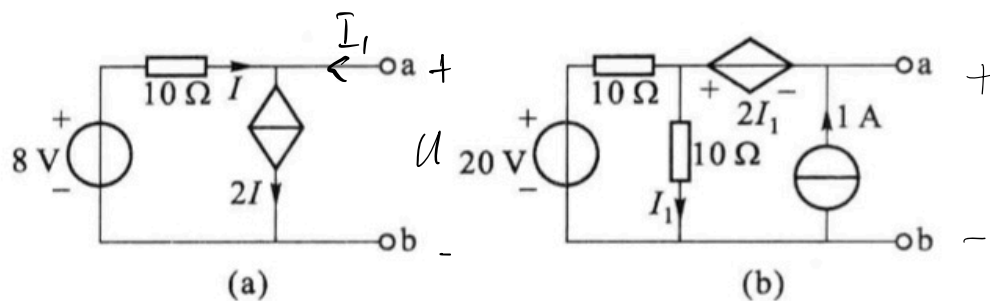
$$\therefore u' = aU_{S1} + bU_{S2} - cI_{S1} = 1.8u \quad 1.8 \text{ 倍}$$

(15~16 学时)

3.3 等效电源定理

3.4 特勒根定理

作业: 3.8, 3.10, 3.13, 3.15, 3.17, 3.18



图题 3.8

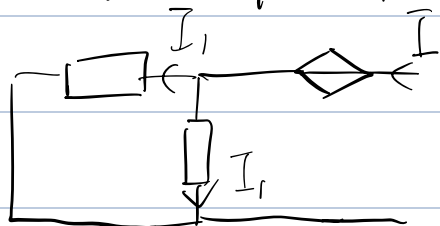
$$3.8 (a) U = -10I + 8. \quad I_1 = 2I - I = I$$

$$U_{oc} = 8V, R_i = -10\Omega$$

$$I_{sc} = -\frac{4}{5}A, G_i = -\frac{1}{10}S$$

$$(b) 10I_1 - 2I_1 = U_{oc} \quad U_{oc} = 8I_1$$

$$20 = 10(I_1 - 1) + 10I_1 \quad \therefore I_1 = 1.5A. \quad U_{oc} = 8 \times 1.5 = 12(V)$$



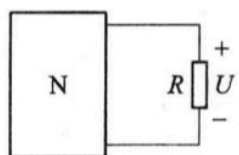
$$I = 2I_1, U = 8I_1$$

$$\therefore R_i = 4\Omega$$

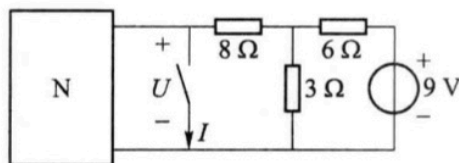
$$U_{oc} = 12V \quad R_i = 4\Omega$$

$$I_{sc} = 3A, G_i = \frac{1}{4}S$$

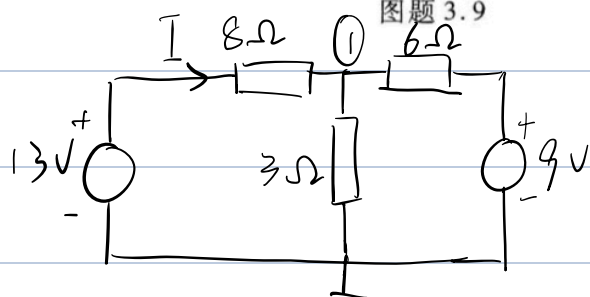
3.10 图中 N 为含独立源电阻网络, 开关断开时量得电压 $U = 13V$, 接通时量得电流 $I = 3.9A$ 。求网络 N 的最简等效电路。



图题 3.9



图题 3.10



$$U_1: \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{8}\right) U_1 = \frac{13}{8} + \frac{9}{6}$$

$$U_1 = 5V$$

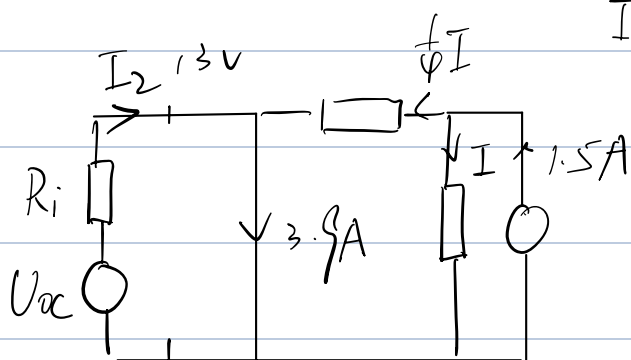
$$I = \frac{13V - 5V}{8\Omega} = 1A$$

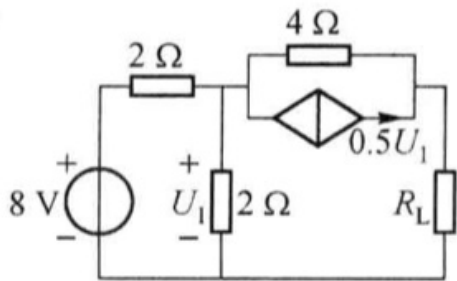
$$\frac{5}{4} I = 1.5A \Rightarrow I = 1.2A$$

$$I_2 = 3.9A - 0.3A = 3.6A$$

$$\therefore \frac{U_{oc}}{R_i} = 3.6A, U_{oc} - 1A \cdot R_i = 13V$$

$$\therefore \begin{cases} U_{oc} = 18V \\ R_i = 5\Omega \end{cases}$$





图题 3.13

$$I + \varphi = U_i$$

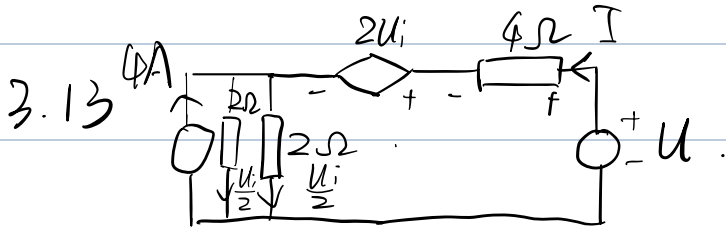
$$U = 3U_i + 4I$$

$$\therefore U = 7I + 12$$

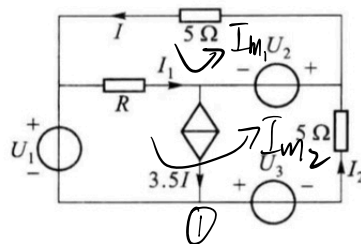
$$\therefore R_i = 7\Omega \quad V_{oc} = 12V$$

$$\therefore R_L = 7\Omega \text{ 时 } \frac{12}{14} A$$

$$P = I^2 R = \left(\frac{12V}{14\Omega}\right)^2 \cdot 7\Omega = \frac{36}{7} W$$



3.15 图示电路, 已知当 $R=2\Omega$ 时, $I_1=5A$, $I_2=4A$ 。求当 $R=4\Omega$ 时 I_1 和 I_2 的值。



图题 3.15

$$I_{m1} = I$$

$$I_{m2} = I_2$$

$$(5+R)I_{m1} - RI_{m2} = U_2 \quad I_{m1} - I_{m2} = I_1$$

$$(5+R)I_{m2} - RI_{m1} = -U_1 - U_2 - U_3$$

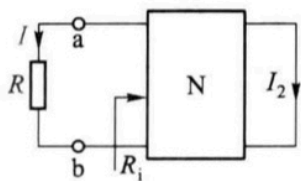
$$\textcircled{1}: I + 3.5I = I_1 + I_2 \quad I = \frac{2}{7}(I_1 + I_2)$$

$$\text{代入 } R=4\Omega$$

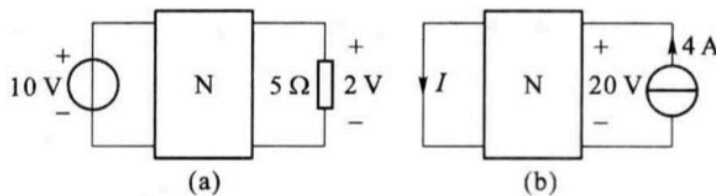
$$\text{解得: } I_1 = \frac{80}{27} A \quad I_2 = \frac{118}{27} A$$

3.17 图示电路中, N 为线性含源电阻网络。已知 $R=0$ 时, $I_2=6\text{ A}$; $R\rightarrow\infty$ 时, $I_2=9\text{ A}$ 。ab 端戴维宁等效电阻为 $R_i=9\ \Omega$ 。求电流 I_2 与电阻 R 的一般关系。

3.18 图示电路中 N 为纯电阻网络, 利用特勒根定理求出电流 I 。



图题 3.17

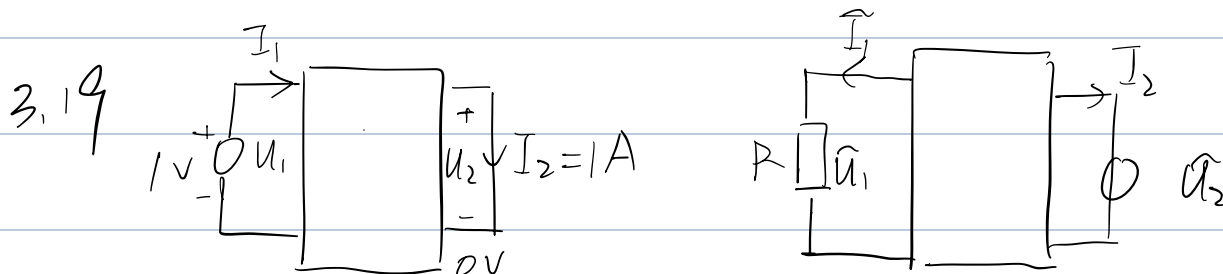


图题 3.18

$$\begin{aligned}
 3.17. \quad I &= \frac{U_{oc}}{9+R} \quad I_2 = aI + bU_{oc} \\
 R \rightarrow +\infty \quad I_2 &= 9\text{ A} \quad \therefore U_{oc} = 81\text{ V} \\
 \begin{cases} 6 = a \cdot \frac{U_{oc}}{9} + b \cdot U_{oc} \\ 9 = b \cdot U_{oc} \end{cases} \\
 \therefore I_2 &= -\frac{1}{3} \frac{81}{9+R} + 9 = \frac{9R+54}{9+R}
 \end{aligned}$$

$$\begin{aligned}
 3.18 \quad 10I - 8 &= 20 \cdot \frac{2}{5} \\
 \therefore I &= 1.6\text{ A}
 \end{aligned}$$

3.19 1.2 1.3 1.5



$$U_1 \tilde{I}_1 + U_2 \tilde{I}_2 = -\tilde{U}_1 \tilde{I}_1 + \tilde{U}_2 \tilde{I}_2$$

$$\therefore 1 \times 1 + 0 = -2R + 5 \times 1$$

$$\therefore 2R = 4(\Omega) \quad R = 2\Omega$$

1.2 图示电路中, 电容 $C=2\text{ F}$, 电容电压 $u_C(t)$ 的波形如图所示。

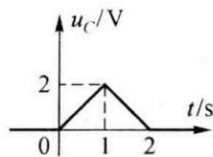
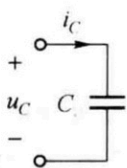
(1) 求电容电流 $i_C(t)$, 并绘出波形图;

(2) 求电容功率表达式, 并绘出功率波形图;

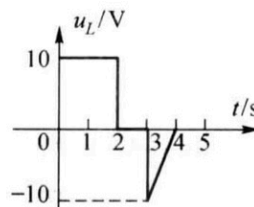
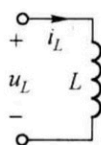
(3) 当 $t=1.5\text{ s}$ 时, 电容是吸收功率还是发出功率? 其值是多少? 电容储能为多少?

1.3 图示电路中, 电感 $L=4\text{ H}$, 电感电压 $u_L(t)$ 的波形如图所示, 已知 $i_L(0)=0$, 试求: (1) 电感电流 $i_L(t)$;

(2) $t=2\text{ s}$ 时电感的储能。

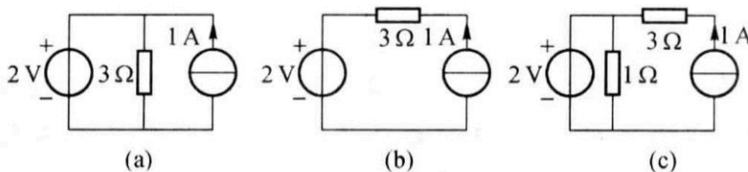


图题 1.2



图题 1.3

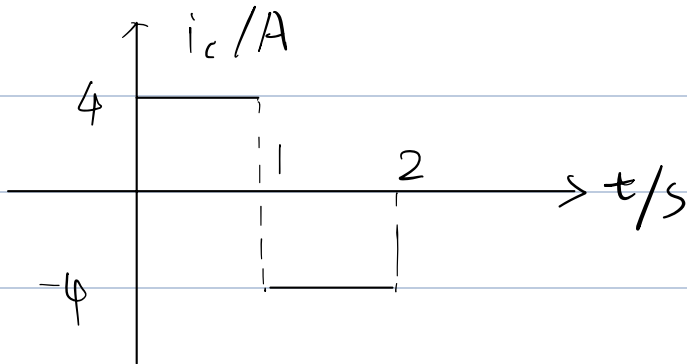
1.4 试分别计算图示三个电路中每个电阻消耗的功率及每个电源所产生的功率。



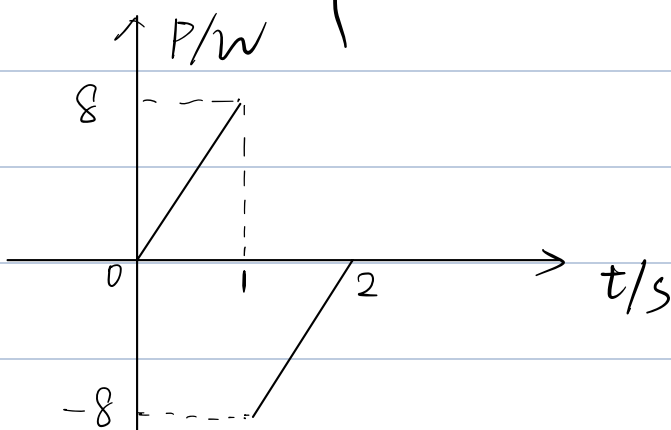
图题 1.4

1.5 计算图示电路电容和电感各自储存的能量。

1.2 (1) $i = \frac{dq}{dt} = C \frac{du}{dt} = \begin{cases} 0, & t < 0 \text{ 或 } t > 2 \\ 4, & 0 \leq t \leq 1 \\ -4, & 1 < t \leq 2 \end{cases} \quad (\text{A})$



$$(2) \quad p = ui = \begin{cases} 0, & t < 0 \text{ 或 } t > 2 \\ 8t, & 0 \leq t \leq 1 \\ 8(t-2), & 1 < t \leq 2 \end{cases}$$



$$(3) \quad t = 1.5 \text{ s} \quad p = 8(1.5 - 2) = -4 \text{ (W)} \quad \text{发出功率 4 W.}$$

$$\text{能量} : \frac{1}{2} CV^2 = \frac{1}{2} \times 2 \times 1^2 = 1 \text{ (J)} \quad \text{储能 1 J}$$

$$1.3. (1) \quad i = \frac{1}{L} \int_{t_0}^t u(\tau) d\tau + 0 = \frac{1}{L} \int_{t_0}^t u(\tau) d\tau$$

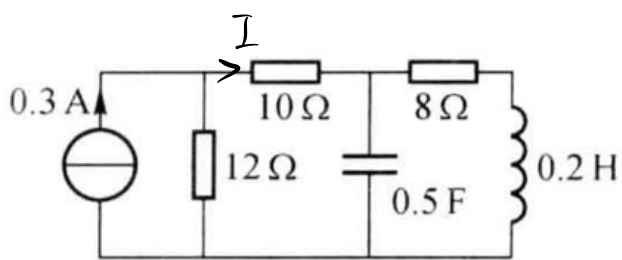
$$0 < t < 2 \quad i = \frac{10}{4} t = 2.5t \text{ (A)}$$

$$2 < t < 3 \quad i = 2.5 \times 2 = 5 \text{ (A)}$$

$$3 < t < 4 \quad i = 5 - \frac{1}{2} \times 1 \times 10 \times (1 - (4-t)^2) \times \frac{1}{4} \\ = 23.75 - 10t + 1.25t^2 \text{ (A)}$$

$$t > 4 \quad i = 5 - \frac{1}{2} \times 10 \times \frac{1}{4} = 3.75 \text{ (A).}$$

$$(2) \quad t = 2. \quad i = 5 \text{ A} \quad \text{能量} : \frac{1}{2} LI^2 = \frac{1}{2} \times 4 \times 5^2 = 50 \text{ (J)}$$



图题 1.5

$$1.5 \quad I = 0.3 \text{ A} \times \frac{12}{12+18} = 0.12 \text{ A}$$

$$\therefore U_C = 0.12 \text{ A} \times 8 \Omega = 0.96 \text{ V}$$

$$\therefore W_C = \frac{1}{2} C U^2 = \frac{1}{2} \times \frac{1}{2} \text{ F} \times (0.96 \text{ V})^2 = 0.2304 \text{ J}$$

$$W_L = \frac{1}{2} L I^2 = \frac{1}{2} \times 0.2 \text{ H} \times (0.12 \text{ A})^2 = 1.44 \times 10^{-3} \text{ J}$$

4.1 4.2

4.3 4.4 4.5 4.6

4.1 已知图示电路中 $u = 100\cos(\omega t + 10^\circ)$ V, $i_1 = 2\cos(\omega t + 100^\circ)$ A, $i_2 = -4\cos(\omega t + 190^\circ)$ A, $i_3 = 5\sin(\omega t + 10^\circ)$ A。试写出电压和各电流的有效值、初相位,并求电压超前于电流的相位差。

4.2 写出下列电压、电流相量所代表的正弦电压和电流(设角频率为 ω):

4.1 u : 有效值 $50\sqrt{2}$ V

初相位 10°

$I_1 \sim \sqrt{2}$ A

$\sim 100^\circ$

$I_2 \sim 2\sqrt{2}$ A

$\sim 10^\circ$

$I_3 \sim \frac{5}{2}\sqrt{2}$ A

$\sim -80^\circ$

u 超前 i : -90°

2 : 0°

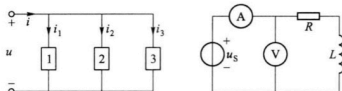
3 : 90°

128 第4章 正弦电流电路

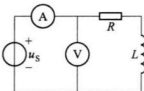
(a) $\dot{U}_u = 10 \angle -10^\circ$ V (b) $\dot{U} = (-6 - j8)$ V

(c) $\dot{I}_u = (0.2 - j20.8)$ A (d) $\dot{I} = -30$ A

4.3 图示电路中正弦电流的频率为 50 Hz 时,电压表和电流表的读数分别为 100 V 和 15 A;当频率为 100 Hz 时,读数为 100 V 和 10 A。试求电阻 R 和电感 L 。

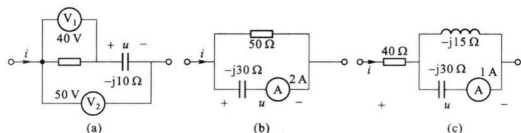


图题 4.1



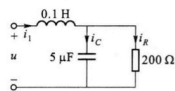
图题 4.3

4.4 图示各电路中已标明电压表和电流表的读数,试求电压 u 和电流 i 的有效值。

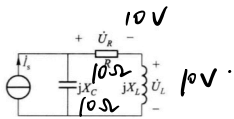


图题 4.4

4.5 在图示电路中已知 $i_R = \sqrt{2}\cos\omega t$ A, $\omega = 2 \times 10^3$ rad/s。求各元件的电压、电流及电源电压 u ,并作各电压、电流的相量图。



图题 4.5



图题 4.6

4.6 已知图示电路中 $U_R = U_L = 10$ V, $R = 10 \Omega$, $X_C = 10 \Omega$, 求 I_S 。

4.2 $U = 10 \cos(\omega t - 10^\circ)$ V

$U = \sqrt{16^2 + 8^2} \cdot \sqrt{2} \cos(\omega t - \arctan \frac{3}{4} - 90^\circ)$

$= 10\sqrt{2} \cos(\omega t - 126.87^\circ)$ (✓)

$i = \sqrt{0.2^2 + 20.8^2} \cos(\omega t - \arctan \frac{20.8}{0.2})$

$= 20.80 \cos(\omega t - 89.45^\circ)$ (A)

$i = 30\sqrt{2} \cos(\omega t + 180^\circ)$ A

4.3 $X_L = 2\pi fL$ 设 50Hz 时为 X_L , 100Hz 时为 $2X_L$.

$$Z_1 = \sqrt{X_L^2 + R^2} = \frac{100}{1.5} = \frac{20}{3}$$

$$Z_2 = \sqrt{4X_L^2 + R^2} = \frac{100}{10} = 10$$

$$\begin{cases} X_L^2 + R^2 = \frac{400}{9} \\ 4X_L^2 + R^2 = 100 \end{cases} \Rightarrow \begin{cases} X_L = \frac{10\sqrt{5}}{3\sqrt{3}} \\ R = \frac{10\sqrt{7}}{3\sqrt{3}} \end{cases}$$

$$\therefore L = \frac{\frac{10\sqrt{5}}{3\sqrt{3}}}{2\pi \times 50} = 0.0137 \text{ H}$$

$$R = 5.092 \Omega$$

4.4 (a) $U = \sqrt{50^2 - 40^2} = 30 \text{ (V)}$

$$\therefore I = \frac{U}{R} = \frac{30}{10} = 3 \text{ (A)}$$

(b) $U = IR = 30\Omega \times 2\text{A} = 60 \text{ V}$

$$i_2 = \frac{U}{R_2} = \frac{60\text{V}}{50\Omega} = 1.2 \text{ (A)}$$

$$\therefore I = \sqrt{1.2^2 + 2^2} \text{ A} = 2.33 \text{ A}$$

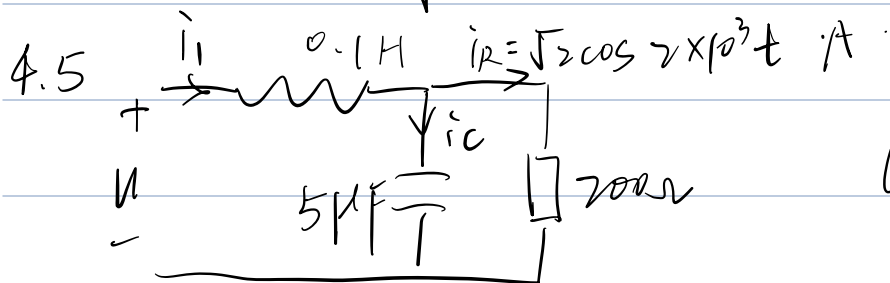
(c) $U_1 = IR = 30\Omega \times 1\text{A} = 30 \text{ V}$

$$i_2 = \frac{U}{R_2} = \frac{30\text{V}}{15\Omega} = 2 \text{ A}$$

$$\therefore I = (2 - 1) \text{ A} = 1 \text{ A}$$

$$\therefore U_2 = IR_3 = 1\text{A} \times 40\Omega = 40 \text{ V}$$

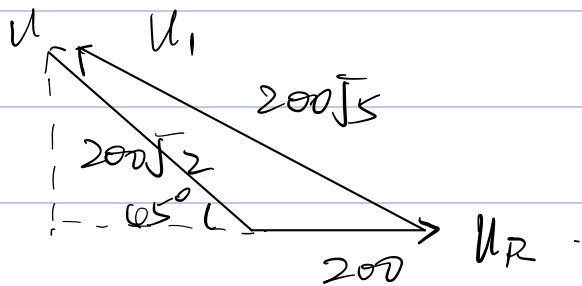
$$\therefore U = \sqrt{30^2 + 40^2} \text{ V} = 50 \text{ V}$$



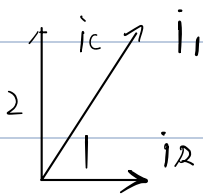
$$U_R = i_R \cdot R$$

$$= 200\sqrt{2} \cos(2 \times 10^3 t) \text{ (V)}$$

$$\dot{U}_R = 200 \angle 0^\circ \text{ (V)}$$



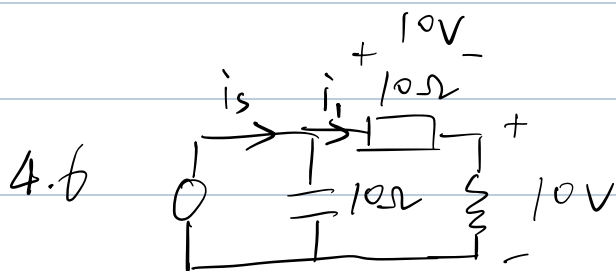
$$\begin{aligned} \dot{i}_C &= \frac{\dot{U}_R}{X_C} = 200 \angle 0^\circ \cdot j\omega C \\ &= 2 \angle 90^\circ \\ &= 2\sqrt{2} \cos(2 \times 10^3 t + 90^\circ) \text{ (A)} \\ \dot{i}_1 &= \dot{i}_C + \dot{i}_R = \sqrt{1^2 + 2^2} \angle \arctan 2 \\ &= \sqrt{5} \angle \arctan 2 \end{aligned}$$



$$\begin{aligned} &= \sqrt{10} \cos(2 \times 10^3 t + 63^\circ) \text{ (A)} \\ \dot{U}_1 &= \dot{i}_1 X_L = \sqrt{5} \angle \arctan 2 \cdot j\omega L \\ &= 200\sqrt{5} \angle (\arctan 2 + 90^\circ) \\ &= 200\sqrt{10} \cos(2 \times 10^3 t + 153^\circ) \text{ (A)} \end{aligned}$$

$$\begin{aligned} U &= U_1 + U_R = 200\sqrt{5} \angle (\arctan 2 + 90^\circ) \\ &\quad + 200 \angle 0^\circ \\ &= 200\sqrt{2} \angle 135^\circ \end{aligned}$$

$$= 400 \cos(2 \times 10^3 t + 135^\circ) \text{ (V)}$$



$$\dot{i}_1 = \frac{10\text{V}}{10\Omega} = 1\text{A}$$

U_L 相位差 U_R 90° .

$$\therefore U_C = 10\sqrt{2} \angle 45^\circ \text{ (V)}$$

$$\dot{i}_C = \frac{U_C}{X_C} = \frac{10\sqrt{2} \angle 45^\circ}{\frac{1}{j} \cdot 10} = \sqrt{2} \angle 135^\circ \text{ (A)}$$

$$\therefore \dot{i}_S = \dot{i}_C + \dot{i}_1 = 1 \angle 90^\circ \text{ (A)} \quad \text{有效值 } 1\text{A}$$

